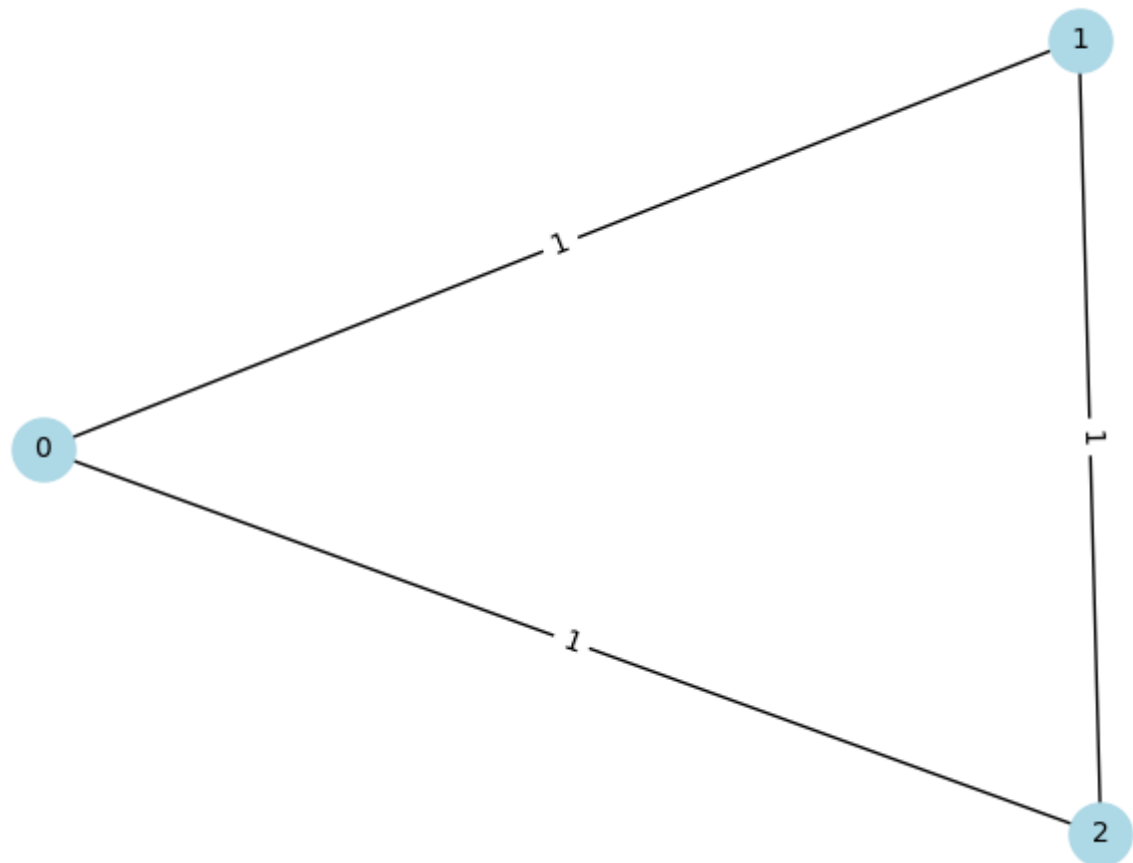


Spanning Tree Protocol

Seed: 3705824578



Consider the network in the figure and assume that:

- The adopted algorithm is the **Spanning Tree Protocol (STP)**.
- BPDUs are formatted as a tuple: (Root ID, Path Cost, Sender ID).
- Tie-breaking preference is: **Lowest Root ID, Lowest Cost, Lowest Sender ID**.
- For simplicity we don't add the ports to the BPDUs, so assume they are generated in node order (from zero on) and they are received in neighbors order
- Every port is initialized in an undefined state, and every port must be assigned for the protocol to converge

Instructions:

Write down the sequence of message exchanges. For every BPDU that causes a state change in the receiver, write:

1. The Sender -> Receiver and the [BPDU] content.
2. The receiver's **Updated Best BPDU**.
3. The receiver's **Updated Port List** using the format [Neighbor:ROLE].

Use the following abbreviations for roles: ROOT, DESI (Designated), BLOC (Blocked), and UNDE (Undefined).

Message Sequence

0 -> 1: [BPDU] (0, 0, 0)

Updated BPDU of 1: (0, 1, 1)

Updated Ports of 1: [0:ROOT, 2:UNDE]

0 -> 2: [BPDU] (0, 0, 0)

Updated BPDU of 2: (0, 1, 2)

Updated Ports of 2: [0:ROOT, 1:UNDE]

1 -> 0: [BPDU] (0, 1, 1)

Updated BPDU of 0: (0, 0, 0)

Updated Ports of 0: [1:DESI, 2:UNDE]

1 -> 2: [BPDU] (0, 1, 1)

Updated BPDU of 2: (0, 1, 2)

Updated Ports of 2: [0:ROOT, 1:BLOC]

2 -> 0: [BPDU] (0, 1, 2)

Updated BPDU of 0: (0, 0, 0)

Updated Ports of 0: [1:DESI, 2:DESI]

2 -> 1: [BPDU] (0, 1, 2)

Updated BPDU of 1: (0, 1, 1)

Updated Ports of 1: [0:ROOT, 2:DESI]

Final Network State

Node 0

Cost: 0

Port State: [1:DESI, 2:DESI]

Node 1

Cost: 1

Port State: [0:ROOT, 2:DESI]

Node 2

Cost: 1

Port State: [0:ROOT, 1:BL0C]